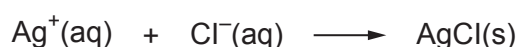


11. Seawater contains a number of dissolved salts. Although composition varies with location, 1 000 cm³ of seawater contains about 20 g of chloride ions, Cl⁻, and about 3 g of sulfate ions, SO₄²⁻.

A student is given a sample of seawater from Rhossili Bay and asked to determine the chloride ion content by volumetric analysis and the sulfate ion content by gravimetric analysis.

- (a) Determination of chloride ion content by volumetric analysis.

The method is similar to an acid-base titration. A silver nitrate solution of known concentration is used to precipitate chloride ions as silver chloride.



The seawater is diluted by a factor of five before it is used in the titration.

The endpoint of this titration is difficult to determine directly, so potassium chromate(VI), K₂CrO₄, is used as an indicator. When all of the chloride ions have been used up, the chromate(VI) ions react with silver ions and produce silver chromate(VI), which forms a red precipitate. The instant a permanent red tinge appears in the solution, the endpoint has been reached.

Volume of diluted seawater in the conical flask = 25.0 cm³

Concentration of silver nitrate solution in the burette = 0.100 mol dm⁻³

Mean titre = 26.40 cm³

- (i) Before starting the titration, the student rinses the burette with silver nitrate solution. Suggest why he does this. [1]

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- (ii) Suggest why the student dilutes the seawater. [1]

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- (iii) Describe how the student should dilute the seawater by a factor of five. [3]

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- (iv) Describe and explain **one** action the student might take just before the endpoint of the titration, to ensure that the volume of silver nitrate added at the endpoint is accurate. [2]

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- (v) Write an ionic equation for the precipitation of silver chromate(VI). [1]

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- (vi) Calculate the mass of chloride ions in $1\,000\text{ cm}^3$ of the original seawater, giving your answer to an **appropriate** number of significant figures. [4]

Mass of chloride ions = g



- (b) Determination of sulfate ion content by gravimetric analysis.

100 cm³ of undiluted seawater and 0.100 mol dm⁻³ barium nitrate solution were used.

The mass of the barium sulfate precipitate was 0.65 g.

You may assume that **all** of the sulfate ions in the seawater were precipitated.

- (i) Describe how the student carried out the gravimetric analysis to find the mass of the barium sulfate precipitated. [5]

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- (ii) Calculate the minimum volume, in cm^3 , of barium nitrate solution needed to precipitate all of the sulfate ions in 100 cm^3 of the seawater.

[3]

Volume = cm^3

- (iii) Suggest why the volume of barium nitrate needed was different to the volume of seawater used.

[1]

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END OF PAPER

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